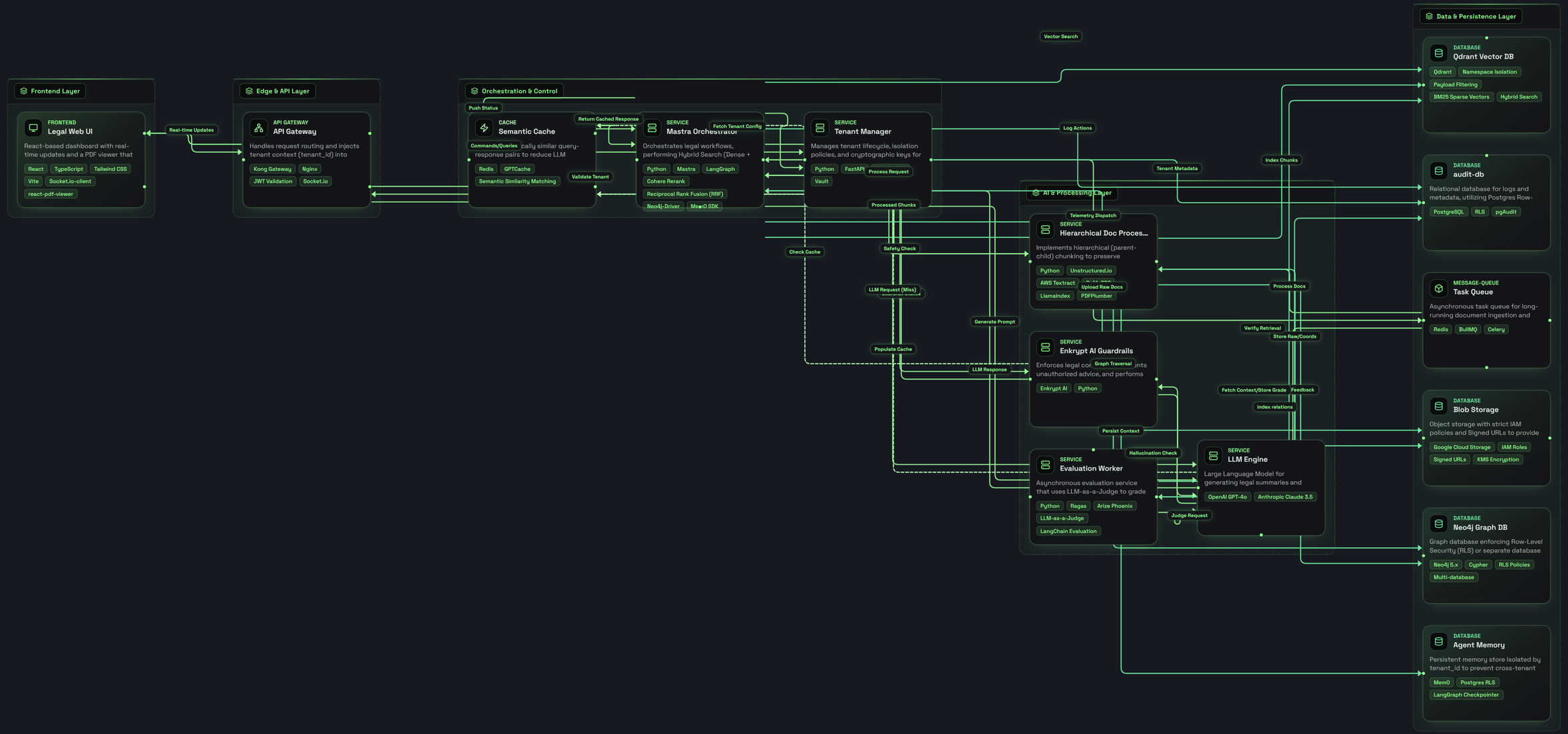
Frontend Layer
FRONTEND
Legal Web UI
React-based dashboard with real-time updates and a PDF viewer that
React
TypeScript
Tailwind CSS
Vite
Socket.io-client
react-pdf-viewer
Real-time Updates
Edge & API Layer
API GATEWAY
API Gateway
Handles request routing and injects tenant context (tenant_id) into
Kong Gateway
Nginx
JWT Validation
Socket.io
Orchestration & Control
Push Status
CACHE
Semantic Cache
Commands/Queries
Redis
GPTCache
Semantic Similarity Matching
Return Cached Response
Validate Tenant
SERVICE
Mastra Orchestrator
Orchestrates legal workflows, performing Hybrid Search (Dense +
Python
Mastra
LangGraph
Cohere Rerank
Reciprocal Rank Fusion (RRF)
Neo4j-Driver
Mem0 SDK
Fetch Tenant Config
SERVICE
Tenant Manager
Manages tenant lifecycle, isolation policies, and cryptographic keys for
Python
FastAPI
Vault
Process Request
Vector Search
Log Actions
Tenant Metadata
Index Chunks
Processed Chunks
Check Cache
Safety Check
LLM Request (Miss)
Populate Cache
Generate Prompt
LLM Response
AI & Processing Layer
Telemetry Dispatch
SERVICE
Hierarchical Doc Proces...
Implements hierarchical (parent-child) chunking to preserve
Python
Unstructured.io
AWS Textract
LlamaIndex
PDFPlumber
Upload Raw Docs
Process Docs
Verify Retrieval
Store Raw/Coords
SERVICE
Enkrypt AI Guardrails
Enforces legal cor nts
Graph Traversal
unauthorized advice, and performs
Enkrypt AI
Python
Fetch Context/Store Grade
Feedback
index relations
Persist Context
Hallucination Check
SERVICE
Evaluation Worker
Asynchronous evaluation service that uses LLM-as-a-Judge to grade
Python
Ragas
Arize Phoenix
LLM-as-a-Judge
LangChain Evaluation
SERVICE
LLM Engine
Large Language Model for generating legal summaries and
OpenAI GPT-4o
Anthropic Claude 3.5
Judge Request
Data & Persistence Layer
DATABASE
Qdrant Vector DB
Qdrant
Namespace Isolation
Payload Filtering
BM25 Sparse Vectors
Hybrid Search
DATABASE
audit-db
Relational database for logs and metadata, utilizing Postgres Row-
PostgreSQL
RLS
pgAudit
MESSAGE-QUEUE
Task Queue
Asynchronous task queue for long-running document ingestion and
Redis
BullMQ
Celery
DATABASE
Blob Storage
Object storage with strict IAM policies and Signed URLs to provide
Google Cloud Storage
IAM Roles
Signed URLs
KMS Encryption
DATABASE
Neo4j Graph DB
Graph database enforcing Row-Level Security (RLS) or separate database
Neo4j 5.x
Cypher
RLS Policies
Multi-database
DATABASE
Agent Memory
Persistent memory store isolated by tenant_id to prevent cross-tenant
Mem0
Postgres RLS
LangGraph Checkpointer